SG No. 222 O_{h}^{2} $Pn\bar{3}n$ [cubic]

* plus set: $+[0,\ 0,\ 0]$
* Wyckoff site: `2a`, site symmetry: `432`

Table 1: Wyckoff bond: `6a@2a`, bond symmetry: `42.2`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,-10,-11,12,-21,22,23,-24]`
4	$[-X,\ 0,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[25,-26,-27,28,41,-42,-43,44]`
5	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[29,-30,-31,32,37,-38,-39,40]`
6	$[0,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[33,-34,-35,36,-45,46,47,-48]`

Table 2: Wyckoff bond: `8b@2a`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,5,9,-14,-19,-24]`
2	$[-X,\ -X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,7,12,-13,-17,-21]`
3	$[-X,\ X,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X,\ -X,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[4,6,11,-16,-18,-23]`
5	$[-X,\ -X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[25,29,33,-38,-43,-48]`
6	$[X,\ X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[26,31,36,-37,-41,-45]`
7	$[X,\ -X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[27,32,34,-39,-44,-46]`
8	$[-X,\ X,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[28,30,35,-40,-42,-47]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-2,13,-14]`
2	$[-X,\ X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[3,-4,-15,16]`
3	$[0,\ X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,-6,23,-24]`
4	$[0,\ -X,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,-8,-21,22]`
5	$[X,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,-10,-19,20]`
6	$[X,\ 0,\ -X]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[11,-12,17,-18]`
7	$[-X,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[25,-26,37,-38]`
8	$[X,\ -X,\ 0]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[27,-28,-39,40]`
9	$[0,\ -X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[29,-30,47,-48]`
10	$[0,\ X,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[31,-32,-45,46]`
11	$[-X,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[33,-34,-43,44]`
12	$[-X,\ 0,\ X]$	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[35,-36,41,-42]`

Table 4: Wyckoff bond: `24d@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Y]$	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-14]`

continued ...

Table 4

No.	vector	center	mapping
2	$[-X, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[2,-13]
3	$[-X, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[3,-15]
4	$[X, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[4,-16]
5	$[Y, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,-24]
6	$[Y, -X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[6,-23]
7	$[-Y, -X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[7,-21]
8	$[-Y, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[8,-22]
9	$[X, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9,-19]
10	$[-X, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[10,-20]
11	$[X, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[11,-18]
12	$[-X, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[12,-17]
13	$[-X, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[25,-38]
14	$[X, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[26,-37]
15	$[X, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[27,-39]
16	$[-X, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[28,-40]
17	$[-Y, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[29,-48]
18	$[-Y, X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[30,-47]
19	$[Y, X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[31,-45]
20	$[Y, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[32,-46]
21	$[-X, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[33,-43]
22	$[X, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[34,-44]
23	$[-X, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[35,-42]
24	$[X, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[36,-41]

Table 5: Wyckoff bond: 24e@2a, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,-2]
2	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[3,-4]
3	$[0, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,-6]
4	$[0, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[7,-8]
5	$[Y, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9,-10]
6	$[Y, 0, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[11,-12]
7	$[Y, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[13,-14]
8	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[15,-16]
9	$[X, 0, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[17,-18]
10	$[-X, 0, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[19,-20]
11	$[0, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[21,-22]
12	$[0, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[23,-24]
13	$[-X, -Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[25,-26]
14	$[X, -Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[27,-28]
15	$[0, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[29,-30]
16	$[0, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[31,-32]
17	$[-Y, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[33,-34]
18	$[-Y, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[35,-36]

continued ...

Table 5

No.	vector	center	mapping
19	$[-Y, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[37,-38]`
20	$[-Y, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[39,-40]`
21	$[-X, 0, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[41,-42]`
22	$[X, 0, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[43,-44]`
23	$[0, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[45,-46]`
24	$[0, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[47,-48]`

Table 6: Wyckoff bond: `48f@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[24]`
25	$[-X, -Y, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25]`
26	$[X, Y, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[26]`
27	$[X, -Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[27]`
28	$[-X, Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[28]`
29	$[-Z, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[29]`
30	$[-Z, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[30]`
31	$[Z, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[31]`
32	$[Z, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[32]`
33	$[-Y, -Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[33]`
34	$[Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[34]`
35	$[-Y, Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[35]`

continued ...

Table 6

No.	vector	center	mapping
36	$[Y, Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[36]
37	$[-Y, -X, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[37]
38	$[Y, X, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[38]
39	$[-Y, X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[39]
40	$[Y, -X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[40]
41	$[-X, -Z, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[41]
42	$[X, -Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[42]
43	$[X, Z, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[43]
44	$[-X, Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[44]
45	$[-Z, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[45]
46	$[-Z, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[46]
47	$[Z, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[47]
48	$[Z, Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[48]

* Wyckoff site: 6b, site symmetry: 42.2

Table 7: Wyckoff bond: 6a@6b, bond symmetry: 42.2

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,-2,-3,4,17,-18,-19,20]
2	$[0, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[5,-6,-7,8,13,-14,-15,16]
3	$[0, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[9,-10,-11,12,-21,22,23,-24]
4	$[-X, 0, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[25,-26,-27,28,41,-42,-43,44]
5	$[0, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[29,-30,-31,32,37,-38,-39,40]
6	$[0, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[33,-34,-35,36,-45,46,47,-48]

Table 8: Wyckoff bond: 12b@6b, bond symmetry: 2.22

No.	vector	center	mapping
1	$[0, X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,-4,18,-19]
2	$[0, -X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[2,-3,-17,20]
3	$[X, 0, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[5,-8,-14,15]
4	$[X, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[6,-7,13,-16]
5	$[X, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[9,-12,21,-24]
6	$[-X, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[10,-11,-22,23]
7	$[0, -X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[25,-28,42,-43]
8	$[0, X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[26,-27,-41,44]
9	$[-X, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[29,-32,-38,39]
10	$[-X, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[30,-31,37,-40]
11	$[-X, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[33,-36,45,-48]
12	$[X, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[34,-35,-46,47]

Table 9: Wyckoff bond: `12c@6b`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,2,-3,-4]`
2	$[X, 0, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,6,-7,-8]`
3	$[0, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9,10,-11,-12]`
4	$[0, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13,14,-15,-16]`
5	$[0, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[17,18,-19,-20]`
6	$[X, 0, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21,22,-23,-24]`
7	$[0, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25,26,-27,-28]`
8	$[-X, 0, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[29,30,-31,-32]`
9	$[0, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[33,34,-35,-36]`
10	$[0, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[37,38,-39,-40]`
11	$[0, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[41,42,-43,-44]`
12	$[-X, 0, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[45,46,-47,-48]`

Table 10: Wyckoff bond: `24d@6b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-19]`
2	$[-Y, -X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,-17]`
3	$[-Y, X, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3,-20]`
4	$[Y, -X, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,-18]`
5	$[X, Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,-14]`
6	$[X, -Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6,-16]`
7	$[-X, -Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[7,-13]`
8	$[-X, Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[8,-15]`
9	$[X, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9,-24]`
10	$[-X, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[10,-22]`
11	$[X, -X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11,-23]`
12	$[-X, -X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[12,-21]`
13	$[-Y, -X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25,-43]`
14	$[Y, X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[26,-41]`
15	$[Y, -X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[27,-44]`
16	$[-Y, X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[28,-42]`
17	$[-X, -Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[29,-38]`
18	$[-X, Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[30,-40]`
19	$[X, Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[31,-37]`
20	$[X, -Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[32,-39]`
21	$[-X, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[33,-48]`
22	$[X, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[34,-46]`
23	$[-X, X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[35,-47]`
24	$[X, X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[36,-45]`

Table 11: Wyckoff bond: `24e@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2]`
2	$[-X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3,-4]`
3	$[0, X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,-6]`
4	$[0, -X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[7,-8]`
5	$[Y, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9,-10]`
6	$[Y, 0, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11,-12]`
7	$[Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13,-14]`
8	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15,-16]`
9	$[X, 0, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[17,-18]`
10	$[-X, 0, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[19,-20]`
11	$[0, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21,-22]`
12	$[0, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[23,-24]`
13	$[-X, -Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25,-26]`
14	$[X, -Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[27,-28]`
15	$[0, -X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[29,-30]`
16	$[0, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[31,-32]`
17	$[-Y, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[33,-34]`
18	$[-Y, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[35,-36]`
19	$[-Y, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[37,-38]`
20	$[-Y, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[39,-40]`
21	$[-X, 0, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[41,-42]`
22	$[X, 0, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[43,-44]`
23	$[0, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[45,-46]`
24	$[0, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[47,-48]`

Table 12: Wyckoff bond: `24f@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-4]`
2	$[0, -X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,-3]`
3	$[Y, 0, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,-8]`
4	$[Y, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6,-7]`
5	$[X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9,-12]`
6	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[10,-11]`
7	$[X, 0, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13,-16]`
8	$[-X, 0, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[14,-15]`
9	$[0, Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[17,-20]`
10	$[0, Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[18,-19]`
11	$[Y, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21,-24]`
12	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22,-23]`
13	$[0, -X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25,-28]`
14	$[0, X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[26,-27]`
15	$[-Y, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[29,-32]`
16	$[-Y, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[30,-31]`
17	$[-X, -Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[33,-36]`

continued ...

Table 12

No.	vector	center	mapping
18	$[X, -Y, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[34,-35]`
19	$[-X, 0, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[37,-40]`
20	$[X, 0, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[38,-39]`
21	$[0, -Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[41,-44]`
22	$[0, -Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[42,-43]`
23	$[-Y, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[45,-48]`
24	$[-Y, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[46,-47]`

Table 13: Wyckoff bond: `48g@6b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[24]`
25	$[-X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25]`
26	$[X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[26]`
27	$[X, -Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[27]`
28	$[-X, Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[28]`
29	$[-Z, -X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[29]`
30	$[-Z, X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[30]`
31	$[Z, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[31]`
32	$[Z, -X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[32]`
33	$[-Y, -Z, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[33]`
34	$[Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[34]`

continued ...

Table 13

No.	vector	center	mapping
35	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[39]`
40	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[40]`
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[41]`
42	$[X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[42]`
43	$[X, Z, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[45]`
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[46]`
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[47]`
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[48]`

* Wyckoff site: `8c`, site symmetry: `.-3.`

Table 14: Wyckoff bond: `8a@8c`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[0, 0, 0]$	`[1,5,9,-25,-29,-33]`
2	$[-X, -X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,7,12,-26,-31,-36]`
3	$[-X, X, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,8,10,-27,-32,-34]`
4	$[X, -X, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6,11,-28,-30,-35]`
5	$[X, X, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[13,17,21,-37,-41,-45]`
6	$[-X, -X, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[14,19,24,-38,-43,-48]`
7	$[X, -X, X]$	$\left[0, \frac{1}{2}, 0\right]$	`[15,20,22,-39,-44,-46]`
8	$[-X, X, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[16,18,23,-40,-42,-47]`

Table 15: Wyckoff bond: `24b@8c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-27]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-28]`
5	$[Z, X, Y]$	$[0, 0, 0]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-32]`
9	$[Y, Z, X]$	$[0, 0, 0]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[14,-38]`

continued ...

Table 15

No.	vector	center	mapping
15	$[Y, -X, Z]$	$[0, \frac{1}{2}, 0]$	`[15,-39]`
16	$[-Y, X, Z]$	$[\frac{1}{2}, 0, 0]$	`[16,-40]`
17	$[X, Z, -Y]$	$[0, 0, \frac{1}{2}]$	`[17,-41]`
18	$[-X, Z, Y]$	$[\frac{1}{2}, 0, 0]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[19,-43]`
20	$[X, -Z, Y]$	$[0, \frac{1}{2}, 0]$	`[20,-44]`
21	$[Z, Y, -X]$	$[0, 0, \frac{1}{2}]$	`[21,-45]`
22	$[Z, -Y, X]$	$[0, \frac{1}{2}, 0]$	`[22,-46]`
23	$[-Z, Y, X]$	$[\frac{1}{2}, 0, 0]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[24,-48]`

* Wyckoff site: `12d`, site symmetry: `-4..`

Table 16: Wyckoff bond: `12a@12d`, bond symmetry: `-4..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, \frac{3}{4}, \frac{1}{4}]$	`[1,4,-41,-44]`
2	$[-X, 0, 0]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[2,3,-42,-43]`
3	$[0, X, 0]$	$[\frac{1}{4}, 0, \frac{3}{4}]$	`[5,8,-37,-40]`
4	$[0, -X, 0]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[6,7,-38,-39]`
5	$[0, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, 0]$	`[9,12,-46,-47]`
6	$[0, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[10,11,-45,-48]`
7	$[0, X, 0]$	$[\frac{3}{4}, 0, \frac{1}{4}]$	`[13,16,-29,-32]`
8	$[0, -X, 0]$	$[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}]$	`[14,15,-30,-31]`
9	$[X, 0, 0]$	$[0, \frac{1}{4}, \frac{3}{4}]$	`[17,20,-25,-28]`
10	$[-X, 0, 0]$	$[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}]$	`[18,19,-26,-27]`
11	$[0, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}]$	`[21,24,-34,-35]`
12	$[0, 0, X]$	$[\frac{1}{4}, \frac{3}{4}, 0]$	`[22,23,-33,-36]`

Table 17: Wyckoff bond: `24b@12d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, \frac{3}{4}, \frac{1}{4}]$	`[1,-4]`
2	$[0, -X, Y]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[2,-3]`
3	$[Y, 0, X]$	$[\frac{1}{4}, 0, \frac{3}{4}]$	`[5,-8]`
4	$[Y, 0, -X]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[6,-7]`
5	$[X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, 0]$	`[9,-12]`
6	$[-X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[10,-11]`
7	$[X, 0, -Y]$	$[\frac{3}{4}, 0, \frac{1}{4}]$	`[13,-16]`
8	$[-X, 0, -Y]$	$[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}]$	`[14,-15]`
9	$[0, Y, -X]$	$[0, \frac{1}{4}, \frac{3}{4}]$	`[17,-20]`
10	$[0, Y, X]$	$[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}]$	`[18,-19]`
11	$[Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}]$	`[21,-24]`
12	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, 0]$	`[22,-23]`
13	$[0, -X, -Y]$	$[0, \frac{1}{4}, \frac{3}{4}]$	`[25,-28]`
14	$[0, X, -Y]$	$[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}]$	`[26,-27]`

continued ...

Table 17

No.	vector	center	mapping
15	$[-Y, 0, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[46,-47]

Table 18: Wyckoff bond: 48c@12d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[31]

continued ...

Table 18

No.	vector	center	mapping
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[32]`
33	$[-Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[33]`
34	$[Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[34]`
35	$[-Y, Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[39]`
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[40]`
41	$[-X, -Z, Y]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[41]`
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[42]`
43	$[X, Z, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[45]`
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[46]`
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[47]`
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[48]`

* Wyckoff site: `12e`, site symmetry: `4..`

Table 19: Wyckoff bond: `12a@12e`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,17,20]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	`[2,3,18,19]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,8,13,16]`
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	`[6,7,14,15]`
5	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[9,12,22,23]`
6	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	`[10,11,21,24]`
7	$[-X, 0, 0]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[25,28,41,44]`
8	$[X, 0, 0]$	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[26,27,42,43]`
9	$[0, -X, 0]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[29,32,37,40]`
10	$[0, X, 0]$	$\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$	`[30,31,38,39]`
11	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	`[33,36,46,47]`
12	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$	`[34,35,45,48]`

Table 20: Wyckoff bond: `24b@12e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2} - x, \frac{1}{4}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[13,-16]`

continued ...

Table 20

No.	vector	center	mapping
8	$[-X, 0, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[17,-20]
10	$[0, Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[46,-47]

Table 21: Wyckoff bond: 48c@12e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[24]

continued ...

Table 21

No.	vector	center	mapping
25	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[48]

* Wyckoff site: 16f, site symmetry: .3.

Table 22: Wyckoff bond: 16a@16f, bond symmetry: .3.

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9]
2	$[-X, -X, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[2,7,12]
3	$[-X, X, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[3,8,10]
4	$[X, -X, -X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[4,6,11]
5	$[X, X, -X]$	$\left[x, x, \frac{1}{2}-x\right]$	[13,17,21]
6	$[-X, -X, -X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[14,19,24]
7	$[X, -X, X]$	$\left[x, \frac{1}{2}-x, x\right]$	[15,20,22]
8	$[-X, X, X]$	$\left[\frac{1}{2}-x, x, x\right]$	[16,18,23]
9	$[-X, -X, -X]$	$[-x, -x, -x]$	[25,29,33]
10	$[X, X, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[26,31,36]
11	$[X, -X, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[27,32,34]
12	$[-X, X, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[28,30,35]
13	$[-X, -X, X]$	$\left[-x, -x, x+\frac{1}{2}\right]$	[37,41,45]
14	$[X, X, X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[38,43,48]
15	$[-X, X, -X]$	$\left[-x, x+\frac{1}{2}, -x\right]$	[39,44,46]
16	$[X, -X, -X]$	$\left[x+\frac{1}{2}, -x, -x\right]$	[40,42,47]

Table 23: Wyckoff bond: `48b@16f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	`[8]`
9	$[Y, Z, X]$	$[x, x, x]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[x, x, \frac{1}{2}-x\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[x, \frac{1}{2}-x, x\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x, x\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[x, x, \frac{1}{2}-x\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, x, x\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[x, \frac{1}{2}-x, x\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[x, x, \frac{1}{2}-x\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[x, \frac{1}{2}-x, x\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-x, x, x\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[24]`
25	$[-X, -Y, -Z]$	$[-x, -x, -x]$	`[25]`
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	`[26]`
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	`[27]`
28	$[-X, Y, Z]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[28]`
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	`[29]`
30	$[-Z, X, Y]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[30]`
31	$[Z, X, -Y]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	`[31]`
32	$[Z, -X, Y]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	`[32]`
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	`[33]`
34	$[Y, -Z, X]$	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	`[34]`
35	$[-Y, Z, X]$	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[-x, -x, x+\frac{1}{2}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[-x, x+\frac{1}{2}, -x\right]$	`[39]`
40	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, -x\right]$	`[40]`
41	$[-X, -Z, Y]$	$\left[-x, -x, x+\frac{1}{2}\right]$	`[41]`
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, -x, -x\right]$	`[42]`
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[-x, x+\frac{1}{2}, -x\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[-x, -x, x+\frac{1}{2}\right]$	`[45]`
46	$[-Z, Y, -X]$	$\left[-x, x+\frac{1}{2}, -x\right]$	`[46]`

continued ...

Table 23

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$\left[x+\frac{1}{2}, -x, -x\right]$	`[47]`
48	$[Z, Y, X]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[48]`

* Wyckoff site: `24g`, site symmetry: `2..`

Table 24: Wyckoff bond: `24a@24g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	`[22,-23]`
13	$[0, -X, -Y]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	`[25,-28]`
14	$[0, X, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[26,-27]`
15	$[-Y, 0, -X]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	`[29,-32]`
16	$[-Y, 0, X]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	`[30,-31]`
17	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	`[33,-36]`
18	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	`[34,-35]`
19	$[-X, 0, Y]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	`[37,-40]`
20	$[X, 0, Y]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[38,-39]`
21	$[0, -Y, X]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[41,-44]`
22	$[0, -Y, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[42,-43]`
23	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	`[45,-48]`
24	$[-Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	`[46,-47]`

Table 25: Wyckoff bond: `24b@24g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	`[5,8]`
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	`[6,7]`
5	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	`[9,12]`
6	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	`[10,11]`
7	$[0, X, 0]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	`[13,16]`
8	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[14,15]`
9	$[X, 0, 0]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[17,20]`

continued ...

Table 25

No.	vector	center	mapping
10	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[18,19]
11	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[21,24]
12	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[22,23]
13	$[-X, 0, 0]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[25,28]
14	$[X, 0, 0]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[26,27]
15	$[0, -X, 0]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[29,32]
16	$[0, X, 0]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[30,31]
17	$[0, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[33,36]
18	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[34,35]
19	$[0, -X, 0]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[37,40]
20	$[0, X, 0]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[38,39]
21	$[-X, 0, 0]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[41,44]
22	$[X, 0, 0]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[42,43]
23	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[45,48]
24	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[46,47]

Table 26: Wyckoff bond: 48c@24g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[26]

continued ...

Table 26

No.	vector	center	mapping
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[48]

* Wyckoff site: 24h, site symmetry: ..2

Table 27: Wyckoff bond: 24a@24h, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{4}, y, y\right]$	[1,-18]
2	$[-Y, -X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$	[2,-20]
3	$[-Y, X, X]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[3,-17]
4	$[Y, -X, X]$	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$	[4,-19]
5	$[-X, Y, X]$	$\left[y, \frac{1}{4}, y\right]$	[5,-15]
6	$[-X, -Y, -X]$	$\left[y, \frac{1}{4}, \frac{1}{2}-y\right]$	[6,-13]
7	$[X, -Y, X]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[7,-16]
8	$[X, Y, -X]$	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$	[8,-14]
9	$[X, -X, Y]$	$\left[y, y, \frac{1}{4}\right]$	[9,-21]
10	$[-X, -X, -Y]$	$\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$	[10,-23]
11	$[X, X, -Y]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[11,-22]
12	$[-X, X, Y]$	$\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[12,-24]
13	$[-Y, -X, X]$	$\left[\frac{3}{4}, -y, -y\right]$	[25,-42]
14	$[Y, X, X]$	$\left[\frac{3}{4}, y+\frac{1}{2}, -y\right]$	[26,-44]
15	$[Y, -X, -X]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[27,-41]
16	$[-Y, X, -X]$	$\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$	[28,-43]
17	$[X, -Y, -X]$	$\left[-y, \frac{3}{4}, -y\right]$	[29,-39]
18	$[X, Y, X]$	$\left[-y, \frac{3}{4}, y+\frac{1}{2}\right]$	[30,-37]
19	$[-X, Y, -X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[31,-40]

continued ...

Table 27

No.	vector	center	mapping
20	$[-X, -Y, X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$	`[32,-38]`
21	$[-X, X, -Y]$	$\left[-y, -y, \frac{3}{4}\right]$	`[33,-45]`
22	$[X, X, Y]$	$\left[y+\frac{1}{2}, -y, \frac{3}{4}\right]$	`[34,-47]`
23	$[-X, -X, Y]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	`[35,-46]`
24	$[X, -X, -Y]$	$\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$	`[36,-48]`

Table 28: Wyckoff bond: `24b@24h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{4}, y, y\right]$	`[1,18]`
2	$[0, -X, X]$	$\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$	`[2,20]`
3	$[0, X, -X]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	`[3,17]`
4	$[0, -X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$	`[4,19]`
5	$[X, 0, X]$	$\left[y, \frac{1}{4}, y\right]$	`[5,15]`
6	$[X, 0, -X]$	$\left[y, \frac{1}{4}, \frac{1}{2}-y\right]$	`[6,13]`
7	$[-X, 0, X]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	`[7,16]`
8	$[-X, 0, -X]$	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$	`[8,14]`
9	$[X, X, 0]$	$\left[y, y, \frac{1}{4}\right]$	`[9,21]`
10	$[-X, X, 0]$	$\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$	`[10,23]`
11	$[X, -X, 0]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	`[11,22]`
12	$[-X, -X, 0]$	$\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$	`[12,24]`
13	$[0, -X, -X]$	$\left[\frac{3}{4}, -y, -y\right]$	`[25,42]`
14	$[0, X, -X]$	$\left[\frac{3}{4}, y+\frac{1}{2}, -y\right]$	`[26,44]`
15	$[0, -X, X]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	`[27,41]`
16	$[0, X, X]$	$\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$	`[28,43]`
17	$[-X, 0, -X]$	$\left[-y, \frac{3}{4}, -y\right]$	`[29,39]`
18	$[-X, 0, X]$	$\left[-y, \frac{3}{4}, y+\frac{1}{2}\right]$	`[30,37]`
19	$[X, 0, -X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	`[31,40]`
20	$[X, 0, X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$	`[32,38]`
21	$[-X, -X, 0]$	$\left[-y, -y, \frac{3}{4}\right]$	`[33,45]`
22	$[X, -X, 0]$	$\left[y+\frac{1}{2}, -y, \frac{3}{4}\right]$	`[34,47]`
23	$[-X, X, 0]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	`[35,46]`
24	$[X, X, 0]$	$\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$	`[36,48]`

Table 29: Wyckoff bond: `48c@24h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, y, y\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$	`[4]`
5	$[Z, X, Y]$	$\left[y, \frac{1}{4}, y\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[y, \frac{1}{4}, \frac{1}{2}-y\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	`[7]`

continued ...

Table 29

No.	vector	center	mapping
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$	[8]
9	$[Y, Z, X]$	$\left[y, y, \frac{1}{4}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y, \frac{1}{4}, \frac{1}{2}-y\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{1}{2}-y\right]$	[14]
15	$[Y, -X, Z]$	$\left[y, \frac{1}{4}, y\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, y, y\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}-y\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{1}{4}, \frac{1}{2}-y, y\right]$	[20]
21	$[Z, Y, -X]$	$\left[y, y, \frac{1}{4}\right]$	[21]
22	$[Z, -Y, X]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-y, y, \frac{1}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, -y, -y\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{3}{4}, y+\frac{1}{2}, -y\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[-y, \frac{3}{4}, -y\right]$	[29]
30	$[-Z, X, Y]$	$\left[-y, \frac{3}{4}, y+\frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[31]
32	$[Z, -X, Y]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[-y, -y, \frac{3}{4}\right]$	[33]
34	$[Y, -Z, X]$	$\left[y+\frac{1}{2}, -y, \frac{3}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[-y, \frac{3}{4}, y+\frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y+\frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[-y, \frac{3}{4}, -y\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{4}, -y, -y\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{3}{4}, y+\frac{1}{2}, y+\frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{3}{4}, y+\frac{1}{2}, -y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[-y, -y, \frac{3}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[y+\frac{1}{2}, -y, \frac{3}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[y+\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$	[48]

* Wyckoff site: 48i, site symmetry: 1

Table 30: Wyckoff bond: `48a@48i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	`[4]`
5	$[Z, X, Y]$	$[z, x, y]$	`[5]`
6	$[Z, -X, -Y]$	$\left[z, \frac{1}{2}-x, \frac{1}{2}-y\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-z, \frac{1}{2}-x, y\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}-z, x, \frac{1}{2}-y\right]$	`[8]`
9	$[Y, Z, X]$	$[y, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-y, z, \frac{1}{2}-x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[y, \frac{1}{2}-z, \frac{1}{2}-x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, \frac{1}{2}-z, x\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2}-z\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[y, \frac{1}{2}-x, z\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, x, z\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[x, z, \frac{1}{2}-y\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, z, y\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}-z, \frac{1}{2}-y\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[x, \frac{1}{2}-z, y\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[z, y, \frac{1}{2}-x\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[z, \frac{1}{2}-y, x\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-z, y, x\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-z, \frac{1}{2}-y, \frac{1}{2}-x\right]$	`[24]`
25	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[25]`
26	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, y+\frac{1}{2}, -z\right]$	`[26]`
27	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[27]`
28	$[-X, Y, Z]$	$\left[-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	`[28]`
29	$[-Z, -X, -Y]$	$[-z, -x, -y]$	`[29]`
30	$[-Z, X, Y]$	$\left[-z, x+\frac{1}{2}, y+\frac{1}{2}\right]$	`[30]`
31	$[Z, X, -Y]$	$\left[z+\frac{1}{2}, x+\frac{1}{2}, -y\right]$	`[31]`
32	$[Z, -X, Y]$	$\left[z+\frac{1}{2}, -x, y+\frac{1}{2}\right]$	`[32]`
33	$[-Y, -Z, -X]$	$[-y, -z, -x]$	`[33]`
34	$[Y, -Z, X]$	$\left[y+\frac{1}{2}, -z, x+\frac{1}{2}\right]$	`[34]`
35	$[-Y, Z, X]$	$\left[-y, z+\frac{1}{2}, x+\frac{1}{2}\right]$	`[35]`
36	$[Y, Z, -X]$	$\left[y+\frac{1}{2}, z+\frac{1}{2}, -x\right]$	`[36]`
37	$[-Y, -X, Z]$	$\left[-y, -x, z+\frac{1}{2}\right]$	`[37]`
38	$[Y, X, Z]$	$\left[y+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$	`[38]`
39	$[-Y, X, -Z]$	$\left[-y, x+\frac{1}{2}, -z\right]$	`[39]`
40	$[Y, -X, -Z]$	$\left[y+\frac{1}{2}, -x, -z\right]$	`[40]`
41	$[-X, -Z, Y]$	$\left[-x, -z, y+\frac{1}{2}\right]$	`[41]`
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, -z, -y\right]$	`[42]`
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, z+\frac{1}{2}, y+\frac{1}{2}\right]$	`[43]`
44	$[-X, Z, -Y]$	$\left[-x, z+\frac{1}{2}, -y\right]$	`[44]`
45	$[-Z, -Y, X]$	$\left[-z, -y, x+\frac{1}{2}\right]$	`[45]`
46	$[-Z, Y, -X]$	$\left[-z, y+\frac{1}{2}, -x\right]$	`[46]`

continued ...

Table 30

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$\left[z+\frac{1}{2}, -y, -x\right]$	[47]
48	$[Z, Y, X]$	$\left[z+\frac{1}{2}, y+\frac{1}{2}, x+\frac{1}{2}\right]$	[48]